Engineering Mechanics 1

Week 9

Yuxia Yuan

Autonomous Aerial Systems

Spatial structures

- Axioms and laws of the plane can be extended to the spatial case

- Statics in 3D:
  - 6 degrees of freedom $u_x, u_y, u_z, M_x, M_y, M_z$
  - 6 equations

$$\vec{M} = \vec{r} \times \vec{F}$$

$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum F_z = 0$$

$$\sum M_x = 0, \quad \sum M_y = 0, \quad \sum M_z = 0$$

- Arrow directions when projecting in 2D ← 3D



outwards



inwards

CC:

2D: $S + I = 3E$

3D: $S + I = 6E$

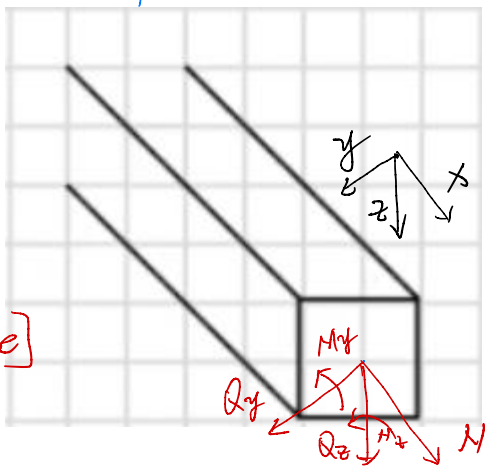
S: S.R. I: I.R. E: No.E

Free cut in 3D

$$\frac{dN}{dx}(x) = -n(x),$$
$$\frac{dQ_z}{dx}(x) = -q_z(x),$$
$$\frac{dQ_y}{dx}(x) = -q_y(x), \text{ [New]}$$

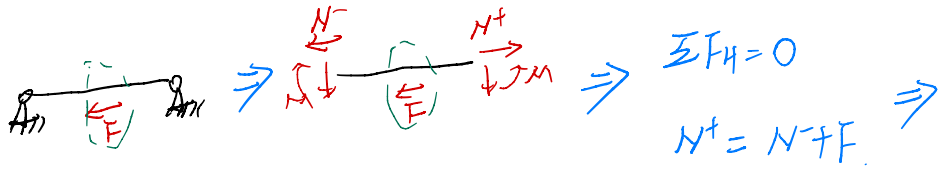
$$\frac{dM_x}{dx}(x) = 0, \text{ [For our course]}$$
$$\frac{dM_y}{dx}(x) = Q_z(x),$$
$$\frac{dM_z}{dx}(x) = -Q_y(x) \text{ [New]}$$

positive section

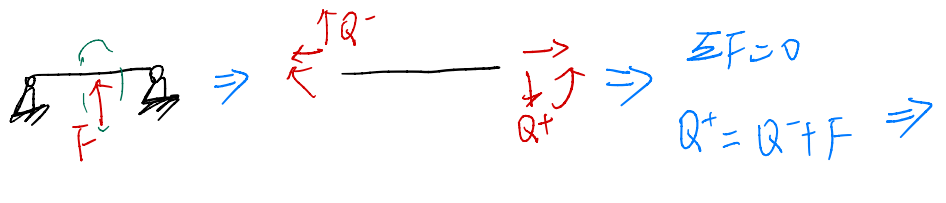


Change direction of forces

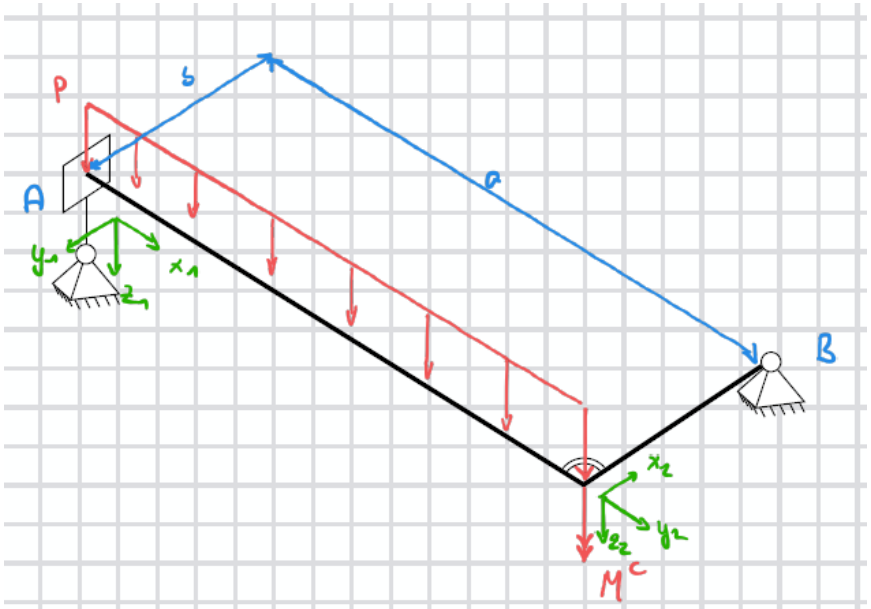
- Normal force



- Shear force.



Exercise 1



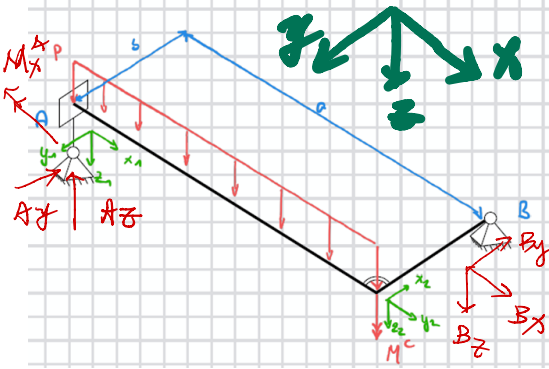
We know

- a = 4 m
- b = 2 m
- p = 6kN/m
- $M_C = 8\ kNm$

Calculate

- Support reactions
- Stress resultants, i.e., N,Q,M

① Support reactions



$\Sigma F_x = 0, \quad B_x = 0$ ①

$\Sigma F_y = 0, \quad A_y + B_y = 0,$ ②

$\Sigma F_z = 0, \quad -A_z + B_z + p \cdot a = 0$ ③

$\Sigma M_{Ax} = 0, \quad -M_x^A - b \cdot B_z = 0$ ④

$\Sigma M_{Ay} = 0, \quad -a \cdot B_z - \frac{a}{2} \cdot pa = 0$ ⑤

$\Sigma M_{Az} = 0, \quad -a \cdot B_y + b \cdot B_x + M_c = 0$ ⑥

$A_y = -2\,KN$

$A_z = 12\,KN$

$M_x^A = 24\,KNm$

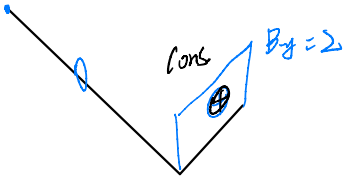
$B_x = 0$

$B_y = 2\,KN$

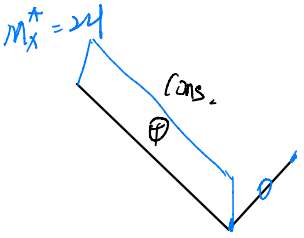
$B_z = -12\,KN$

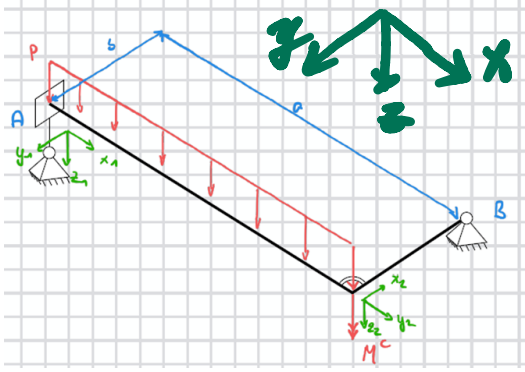
N [KN]

support Ax

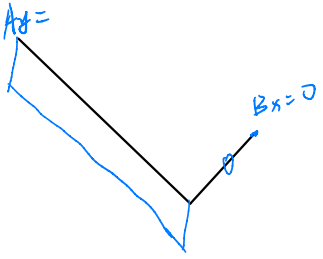


Mx [KNm]

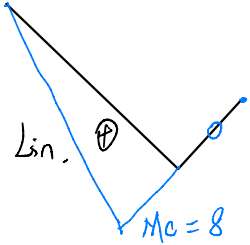




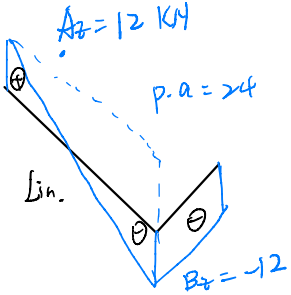
Qy [KN]



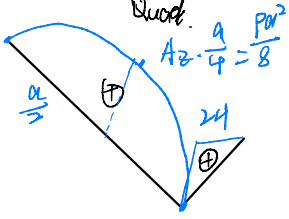
Mz [KNm]



Qz [KN]



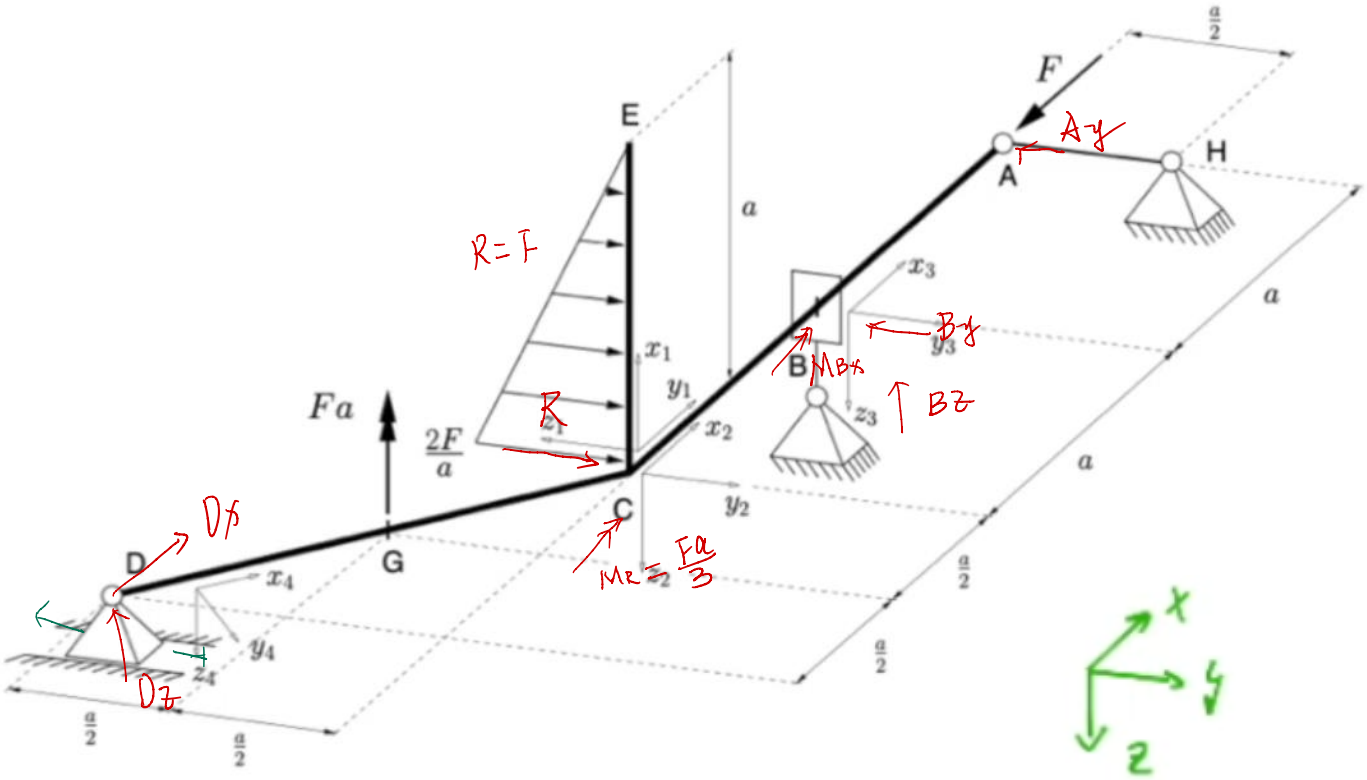
My [KNM]



$$\frac{dM_z}{dx} = -Q_y$$

$$\frac{dM_y}{dx} = Q_z$$

Exercise 2



We know
- a, F

Calculate
- Qy, Mz

① Support reactions

$$\Sigma F_x = 0, \quad D_x - F = 0 \qquad ①$$

$$\Sigma F_y = 0, \quad R - A_y - B_y = 0$$

$$\Sigma F_z = 0, \quad D_z + B_z = 0$$

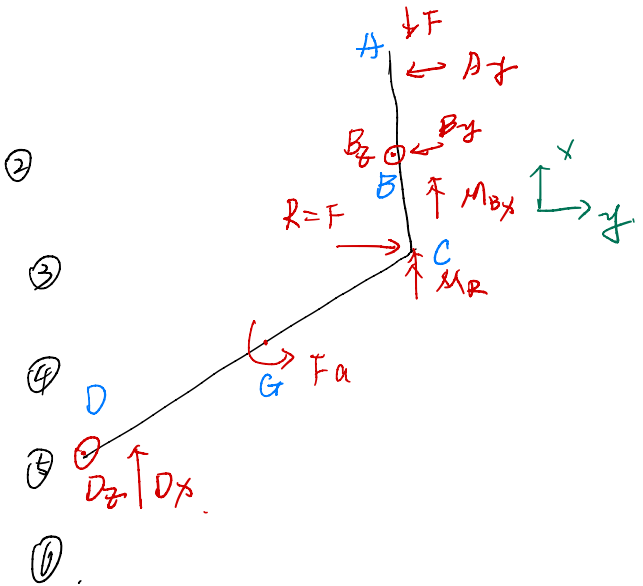
$$\Sigma M_z^B = 0, \quad aA_y + a \cdot R + Fa - a \cdot D_x = 0$$

$$\Sigma M_y^D = 0, \quad 2a \cdot B_z$$

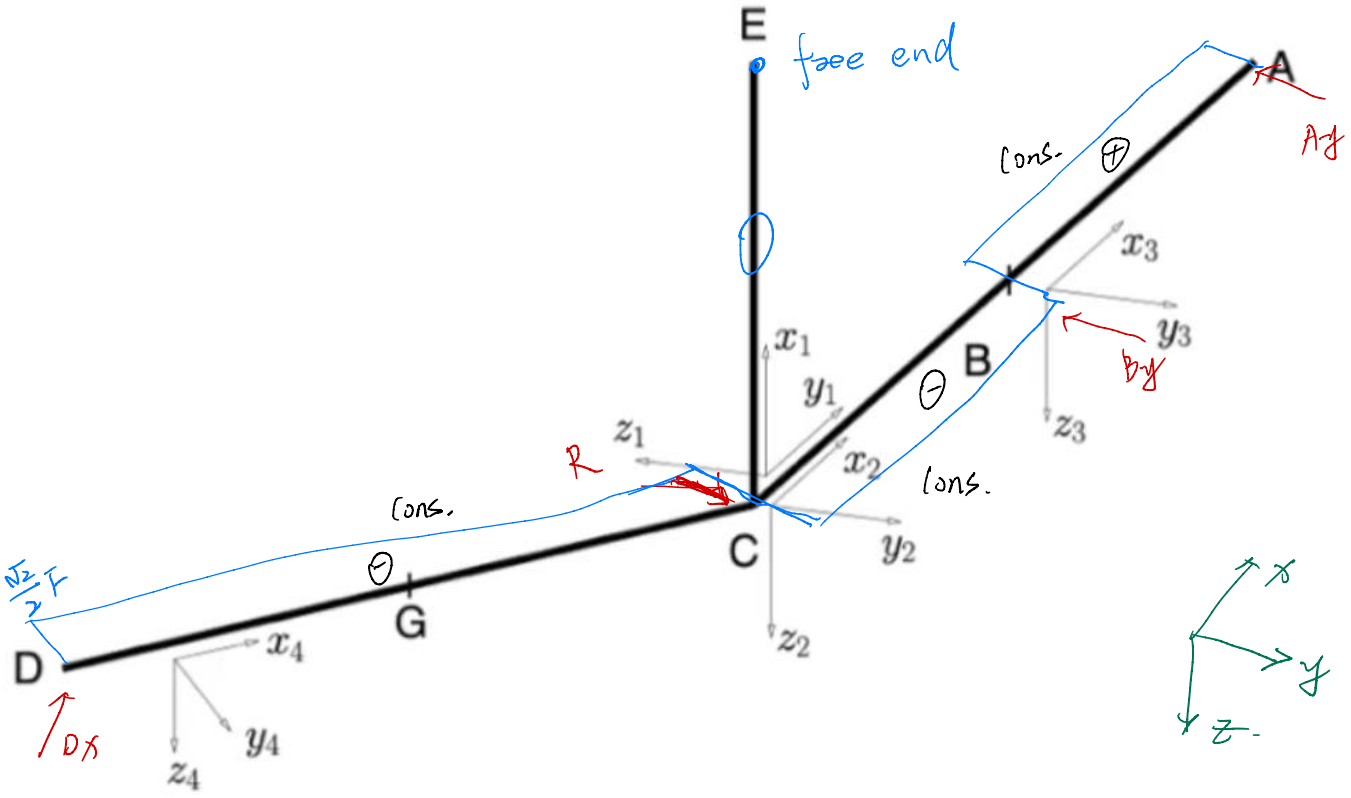
$$\Sigma M_x^C = 0, \quad M_R + M_{Bx} + a \cdot D_z$$

$$\begin{cases} D_x = F \\ A_y = F \\ B_y = 2F \\ B_z = 0 \\ D_z = 0 \\ M_x^B = -\frac{a}{3}F \end{cases}$$

projection

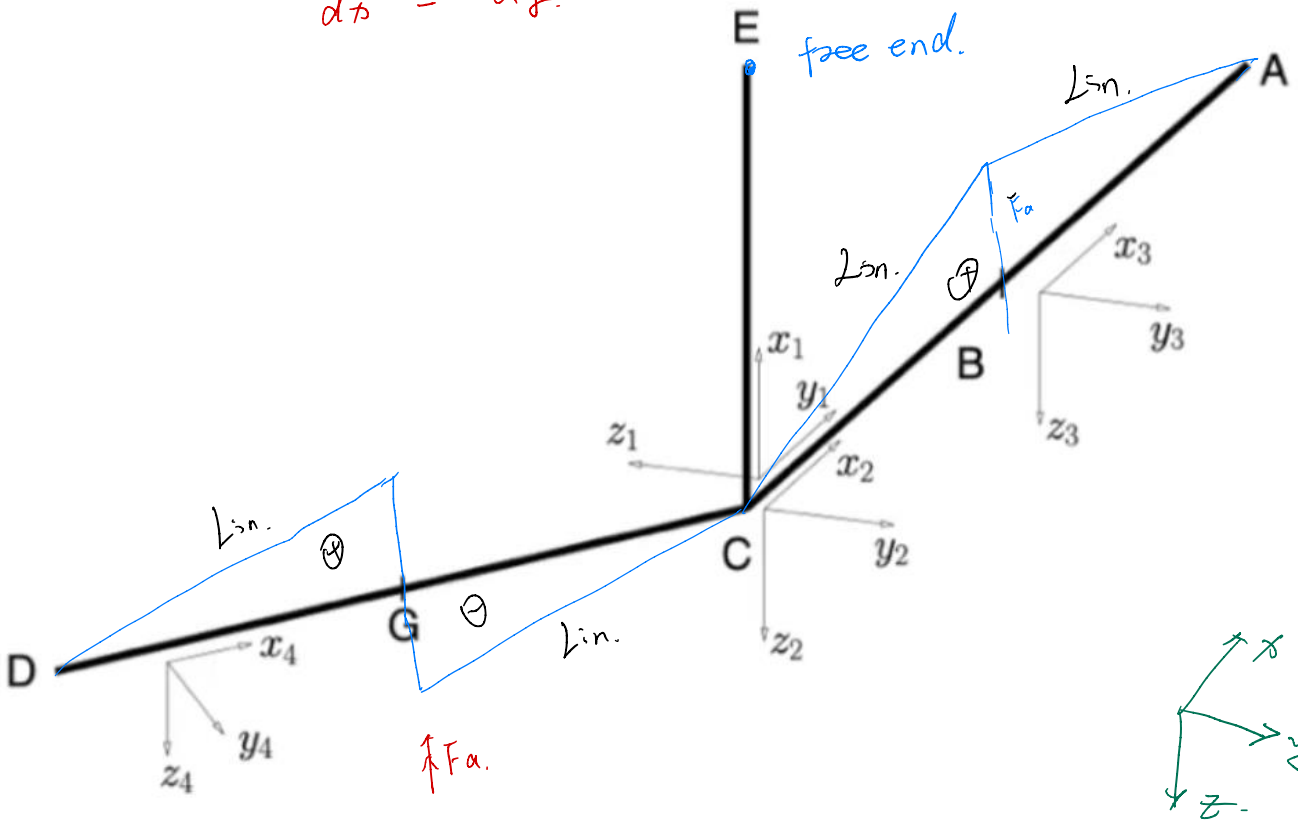


Qy.


E
free end
A
Ay
cons.
θ
x_3
y_3
By
x_1
y_1
z_1
R
B
θ
z_3
x_2
cons.
cons.
y_2
C
θ
G
D
x_4
y_4
z_4
z_2
Dx
x
y
z

M_z

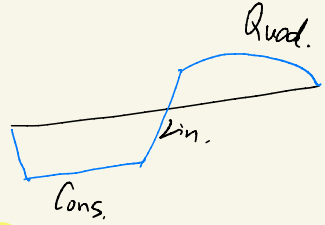
$$\frac{dM_z}{dz} = -Q_y.$$



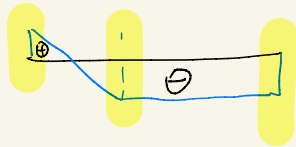
Diagrams in the Exam.

→ What do you have to draw

1) Polynomial order

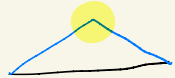


2) Sign of stress

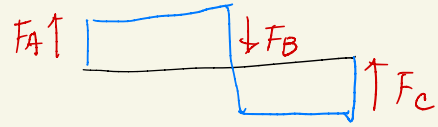


3) start and end of every section.

4) discontinuities



5) reasons for offsets (jumps)



6) in quadratic and cubic cases, extrem values and locations

